Mock exam 1: answers

Section A: Multiple choice questions

1 [1] easy

Which of these is a magnetic material?

(a) **Nickel**
(b) Wood
(c) Glass
(d) Plastic

Answer (a) Nickel

Iron, nickel and cobalt are magnetic materials. Wood, glass and plastic are not attracted by a magnet.

Read in the chapter: 4.1 Magnetic and Non-magnetic Materials

Back to question 1

2 [1] easy

Magnets that are found in nature are called

(a) artificial magnets
(b) **lodestones**
(c) compasses
(d) iron filings

Answer (b) lodestones

Lodestones are natural magnets. Magnets made by people are artificial magnets.

Read in the chapter: Introduction

Back to question 2

3 [1] easy

A bar magnet is placed on a heap of iron filings and lifted. Where do most of the filings stick?

(a) In the middle of the magnet

(b) Equally all over the magnet

(c) **Near the two ends of the magnet**

(d) Only on one end of the magnet

Answer (c) Near the two ends of the magnet

The two ends are the poles. A magnet attracts the most at its poles.

Read in the chapter: 4.2 Poles of Magnet

Back to question 3

4 [1] easy

A bar magnet hangs freely from a thread. Along which direction does it come to rest?

(a) Any direction

(b) East-west

(c) It never comes to rest

(d) **North-south**

Answer (d) North-south

A freely suspended magnet always rests along the north-south direction.

Read in the chapter: 4.3 Finding Directions

Back to question 4

5 [1] easy

The South pole of one magnet is brought near the South pole of another magnet. The two poles will

(a) repel each other

(b) attract each other

(c) not affect each other

(d) first attract and then repel

Answer (a) repel each other

South and South are like poles. Like poles repel each other.

Read in the chapter: 4.4 Attraction and Repulsion between Magnets

Back to question 5

6 [1] medium

Assertion (A): A magnet can move a compass needle even when a sheet of cardboard is kept between them.
Reason (R): The magnetic effect can act through non-magnetic materials.

(a) Both A and R are true, and R explains A.

(b) Both A and R are true, but R does not explain A.

(c) A is true, but R is false.

(d) A is false, but R is true.

Answer (a) Both A and R are true, and R explains A.

Cardboard is a non-magnetic material. The magnetic effect passes through it, so the needle still moves.

Read in the chapter: 4.4 Attraction and Repulsion between Magnets

Back to question 6

Section B: Very short answer questions

7 [2] easy

Name the two poles of a magnet. Can a magnet have only one pole?

Answer The North pole and the South pole. No, a magnet cannot have only one pole.

Poles always exist in pairs, even in the smallest piece of a magnet.

Read in the chapter: 4.2 Poles of Magnet

Back to question 7

8 [2] easy

Write any two things that you should not do to a magnet.

Answer Any two of these: do not heat it, do not drop it, do not hammer it, do not keep it near mobile phones or remote controls.

These are the rules for treating magnets with care.

Read in the chapter: 4.5 Fun with Magnets

Back to question 8

Section C: Short answer questions

9 [3] medium

Meena brings a magnet near an iron nail, a plastic scale and a glass tumbler.
(a) Which object sticks to the magnet?
(b) Which objects are made of non-magnetic materials?
(c) What is a magnetic material?

Answer (a) The iron nail. (b) The plastic scale and the glass tumbler. (c) A material that is attracted towards a magnet.

1. Iron is a magnetic material, so the nail sticks.
2. Plastic and glass are not attracted towards a magnet, so they are non-magnetic.
3. Materials that are attracted towards a magnet are called magnetic materials.

Read in the chapter: 4.1 Magnetic and Non-magnetic Materials

Back to question 9

10 [3] medium

You are standing in a playground with a magnetic compass. Write three steps to find the directions with it.

Answer Keep the compass still, wait for the needle to rest, then turn the box until the dial matches the needle.

1. Keep the compass at the place where you want to know the directions.
2. Wait till the needle comes to rest. It rests along the north-south direction.
3. Gently rotate the box until north and south on the dial line up with the needle. Now read the directions from the dial.

Read in the chapter: 4.3 Finding Directions

Back to question 10

Section D: Long answer question

11 [5] easy

Arun places a bar magnet on a few round pencils. He brings a second bar magnet near it, without touching.

(a) What happens when he brings a North pole near the North pole of the first magnet?

(b) What happens when he brings a South pole near the North pole of the first magnet?

(c) What are these two effects called?

(d) He now replaces the magnet on the pencils with an iron bar. What will he see when he brings each pole of his magnet near it? What does this tell him?

Answer (a) The magnet on the pencils moves away. (b) It moves closer. (c) Repulsion and attraction. (d) The iron bar is attracted by both poles, so it is not a magnet.

1. (a) North and North are like poles. Like poles repel, so the magnet rolls away. (1 mark)
2. (b) South and North are unlike poles. Unlike poles attract, so the magnet rolls closer. (1 mark)
3. (c) The push is called repulsion and the pull is called attraction. (1 mark)
4. (d) Both ends of an iron bar are attracted by both poles of a magnet. It is never repelled. Only a magnet can be repelled, so the iron bar is not a magnet. (2 marks)

Read in the chapter: 4.4 Attraction and Repulsion between Magnets

Back to question 11

Section E: Case-based question

12 [4] medium

Read the passage and answer the questions.
Reshma sets up three games at her school fair. In Game 1, steel balls lie inside a maze on a cardboard tray, and players move a magnet below the tray. In Game 2, a steel paper clip lies in a glass of water. In Game 3, two matchbox cars each carry a bar magnet, with the South poles facing each other.

(i) In Game 1, why do the steel balls move although the magnet does not touch them?
Answer The magnetic effect acts through the cardboard, which is a non-magnetic material.

(ii) In Game 2, how can a player take out the clip without wetting the fingers or the magnet?
Answer Hold the magnet against the outside of the glass near the clip and slide it up. The clip follows the magnet.

(iii) In Game 3, what happens when the two cars are brought close and let go? Why?
Answer The cars move away from each other, because like poles repel.

(iv) What one change will make the two cars rush towards each other?
Answer Turn one car (or its magnet) around so that a North pole faces a South pole. Unlike poles attract.

Each part carries 1 mark.

Read in the chapter: 4.5 Fun with Magnets

Back to question 12